

Exposure timing is a determinant of fine particulate matter (PM_{2.5}) pulmonary, vascular, and metabolic toxicity in male mice

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Abstract

Exposure to fine particulate matter (PM_{2.5}) air pollution is associated with an increased cardiometabolic disease risk. However, although exposure timing is recognized as an important toxicity determinant, the chronotoxicity of PM_{2.5} exposure is less explored. To evaluate whether PM_{2.5} sensitivity depends on exposure timing, adult male C57/BL6 mice were exposed to concentrated PM_{2.5} (CAP, 6 h/day, 30 days) either during the inactive (light) phase (Zeitgeber time, ZT1 to 7) or the active (dark) phase (ZT17 to 23). Metabolic health was assessed by measuring body weight, fasting blood glucose, and plasma insulin levels. Oxidative stress and inflammatory responses in aortas and lungs were examined by quantitative real-time (qRT)-PCR, and pulmonary and circulating redox changes were measured calorimetrically. Although CAP exposure during the active phase increased blood glucose levels, inactive-phase exposure more profoundly increased antioxidant enzyme and inflammatory mRNA abundance in lungs and aortas. Inactive-phase exposure also intensified pulmonary and systemic lipid peroxidation and increased the depletion of circulating nitric oxide and lung glutathione. Examining the ability to protect against CAP-induced pulmonary oxidative stress, we found that the pulmonary antioxidant enzyme mRNA expression showed circadian rhythmicity that peaked during the active phase. These data suggest that a superior pulmonary antioxidant defense potential during the active phase could contribute to the protection against the PM_{2.5}-induced vascular and pulmonary effects but not its impact on glucose homeostasis. Our study identified that exposure sensitivity depends on the exposure timing, which is of significance as it informs on timely susceptibility windows that could help to mitigate PM_{2.5} toxicity.

Keywords: air pollution; chronotoxicity; fine particulate matter (PM_{2.5}); pulmonary oxidative stress; vascular inflammation

Air pollution exposure is linked to an increase in cardiovascular and metabolic disease (EPA 2025; NIEHS 2025). Although overall air pollution exposure has been identified as a major health burden, inhalation of fine particulate matter (PM_{2.5}) seems to be specifically harmful to cardiovascular and metabolic health (Brook et al. 2010; Pearson et al. 2010; Pope et al. 2015; Rao et al. 2015; Rajagopalan et al. 2024; Munzel et al. 2025). In humans, subacute PM_{2.5} exposure of a cohort of male and female subjects decreased insulin sensitivity (Brook et al. 2013). In addition, several population studies identified an increase in cardiovascular events after a short-term rise in ambient PM_{2.5} levels (Pope et al. 2006; O'Donnell et al. 2011; Samoli et al. 2014; Stafoggia et al. 2022). Other studies have associated long-term exposures to elevated PM_{2.5} levels with the development of diabetes (Chen et al. 2013; Xi et al. 2026) and cardiovascular disease (Miller et al. 2007; Bauer et al. 2010; Chan et al. 2015; Jin et al. 2022) in mixed-sex or female cohorts. In male mice, short-term exposures to concentrated PM_{2.5} (CAP) impaired vascular repair mechanisms and exacerbated glucose intolerance in diet-induced obesity and

light-induced circadian dyssynchrony (Haberzettl et al. 2012, 2016b, 2018; Ribble et al. 2023). Long-term exposures have been found to induce endothelial dysfunction and insulin resistance and to accelerate atherosclerosis in male mice (Sun et al. 2005; Kampfrath et al. 2011; Xu et al. 2011). In contrast, female mice were protected against the vascular and metabolic toxicity of short-term exposure to concentrated PM_{2.5} (Kitching et al. 2025).

Although the health impact of exposure duration (i.e. acute vs. chronic) has been extensively examined, it has not been investigated whether PM_{2.5} toxicity depends on the time of the day the exposure occurs. Daily variations in the murine sensitivity to PM_{2.5} constituents such as metals, but also other toxins such as bromobenzene and acrylamide (Rana et al. 2007; Miura et al. 2013; Yoshioka et al. 2017, 2018; Wang et al. 2023), suggest that exposure timing is a crucial determinant of toxicity. However, even though Kerr (1973) emphasized the importance of exposure timing in a controlled air pollution study, PM_{2.5} chronotoxicity has not been greatly considered, although it is likely that daily windows that increase the susceptibility to PM_{2.5} exist.

Identifying sensitive exposure windows is of importance, as currently emission regulations that have reached their limits and exposure avoidance, which are deemed difficult to implement, are the only approaches employed to mitigate against the harmful effects of air pollution inhalation. Therefore, recognizing time-of-day sensitivity to air pollution could help to encourage behavioral changes to avoid air pollution exposure at certain susceptible times. Such timely exposure prevention would be easier to obey, which could help to lessen the global burden of air pollution on cardiovascular and metabolic health.

Although the specific mechanism(s) by which PM_{2.5} exposure impairs cardiovascular and metabolic health is still not completely understood, one important contributor to the cardiovascular and metabolic toxicity of PM_{2.5} seems to be the induction of oxidative stress (Kampfthaler et al. 2011; Haberzettl et al. 2014, 2016b, 2018; Li et al. 2016; Strak et al. 2017; Riggs et al. 2020; Kurlawala et al. 2023; Ribble et al. 2023; Kitching et al. 2025; Munzel et al. 2025). Interestingly, increasing evidence indicates that antioxidant defense mechanisms and the potential to avert oxidative stress show diurnal patterns (Blanco et al. 2007; Jang et al. 2011; Xu et al. 2012; Wilking et al. 2013; Pekovic-Vaughan et al. 2014). Consequently, it is possible that diurnal changes in the capacity to deal with air pollution-induced oxidative stress could impact PM_{2.5} toxicity. Our previous studies showed that in mice, exposure to concentrated PM_{2.5} during the inactive (light) phase induced inflammation and oxidative stress in the blood vessels by inducing pulmonary oxidative stress without impacting glucose homeostasis (Haberzettl et al. 2016b, 2018; Ribble et al. 2023). However, we have neither tested whether other exposure windows (i.e. exposure during the active (dark) phase) impact cardiovascular and metabolic health nor examined whether PM_{2.5} during different activity phases induce different pulmonary, vascular, or systemic responses. In the current study, we examined whether exposure during either the inactive (light) or the active (dark) phase modifies PM_{2.5} responses in lungs and blood vessels. Although we have not observed that short-term PM_{2.5} exposure (9 to 30 days) during the inactive phase changes body weight or glucose homeostasis independent of diet-induced obesity or light-induced circadian dyssynchrony (Haberzettl et al. 2012, 2016b, 2018; Ribble et al. 2023), we measured metabolic parameters such as body weight, blood glucose, and plasma insulin levels to examine whether active-phase exposure induces such metabolic changes. Finally, we examined whether the time-of-day PM_{2.5} susceptibility (chronotoxicity) is associated with circadian differences in the pulmonary antioxidant defense.

Materials and methods

Animal experiments

Adult male C57/BL6 mice were purchased from the Jackson Laboratories and acclimated to our housing conditions (ZT = Zeitgeber in circadian time, ZT0 = beginning of light cycle, ZT12 = beginning of the dark cycle, 12 h light [ZT0 to 12]: 12 h dark [ZT12 to 24]) for 1 wk. While our previous study indicates that female mice are not susceptible to PM_{2.5}-cardiotoxicity (Kitching et al. 2025), epidemiological studies indicate a higher risk for air pollution exposure-associated cardiovascular disease in women (Adar et al. 2013; Bell et al. 2015; Su et al. 2016; Cheng et al. 2019; Curto et al. 2019). However, the increased cardiovascular disease risk seems to depend on age. Women at reproductive age (18 to 45 yr of age) were at lower risk than men of similar age (Xie et al. 2018; Curto et al. 2019; Wang et al. 2020). Based on previous

findings that young female mice are protected against cardiovascular PM_{2.5} toxicity (Kitching et al. 2025), we use only male mice. Similarly, Qin et al. (2018) and Yan et al. (2020) found that oropharyngeal aspiration of PM_{2.5} for 4 wks neither induced cardiovascular nor hepatic toxicity in adult female mice, whereas it induced cardiac fibrosis and disturbed hepatic lipid metabolism in juvenile (4-wk-old) and older (10-mo-old) female mice. After long-term (24 wks) PM_{2.5} exposure, female mice showed higher hepatic lipid accumulation and were more likely to develop systemic insulin resistance than male mice (Li et al. 2020). In contrast, in a study using adult (age: 8 to 10 wks) female and male mice, Liu et al. (2021) showed that female mice were protected against endothelial progenitor cell depletion after intranasal instillation of PM_{2.5} that was associated with a lower systemic inflammatory and oxidative stress response in females. A rodent study by Wang et al. (2022) showed that pulmonary, systemic, and vascular responses to PM_{2.5} treatment were similar in male and female mice. Regardless, due to the existing evidence that adult female mice seem to be protected against cardiovascular and metabolic toxicity of short-term PM_{2.5} exposure, only male mice were used.

As described previously (Haberzettl et al. 2012, 2016a, 2016b, 2018; Ribble et al. 2023), mice were exposed for 30 consecutive days (6 h per day) to concentrated PM_{2.5} (CAP) or inhaled HEPA-filtered air (control) using a whole-body inhalation system (versatile aerosol concentration enrichment system, VACES). To test for daily variations in the susceptibility to PM_{2.5} (chronotoxicity, Fig. 1A), mice were exposed during either the inactive (light) phase (ZT1 to 7, Experiment 1) or the active (dark) phase (ZT17 to 23, Experiment 2).

The results from the gravimetric filter weight analysis (ZT1 to 7, inactive phase: ambient PM_{2.5}: 12.75 ± 2.11 µg/m³, HEPA-filtered air: 1.21 ± 0.35 µg/m³, CAP: 97.97 ± 11.18 µg/m³; ZT 17 to 23, active phase: ambient PM_{2.5}: 10.58 ± 1.61 µg/m³, HEPA-filtered air: 2.06 ± 0.52 µg/m³; CAP: 72.13 ± 10.01 µg/m³) indicate that the VACES concentrates ambient PM_{2.5} by about 7-fold. No significant differences in CAP concentrations (P = 0.160) occurred between inactive and active-phase exposure. Although concentrated, these experimental particle concentrations are relevant as they reflect frequent human exposure scenarios to ambient PM_{2.5} levels of 5 to 100 µg/m³ (Rajagopalan and Landrigan 2021). Moreover, the mouse exposure bears directly on the impact of PM_{2.5} exposure on human health as mice inhale real-world particles that are collected from downtown Louisville air (O'Toole et al. 2010; Haberzettl et al. 2012). Daily measurements of real-time mass concentration (Fig. 1B, 30-day mean CAP concentration ZT1 to 7: 106.96 ± 11.01 µg/m³, ZT17 to 23: 84.94 ± 9.06 µg/m³, P = 0.128) using a nephelometer (DataRAM 4) confirmed the values of the particle dose obtained from the gravimetric filter measurements. In addition, the measured mass diameter range of 0.39 to 2.25 µm (Fig. 1C) verifies that mice were exposed to particles with a mean diameter of <2.5 µm.

To test whether pulmonary, vascular, and systemic PM_{2.5} toxicity is dependent on the exposure timing, mice were randomly assigned to experimental groups that inhale air (n = 30) or CAP (n = 30) during either the inactive (Experiment 1) or active (Experiment 2) phase. Data obtained from air-inhaling mice in Experiment 1 and Experiment 2 were pooled. An additional set of air-inhaling mice (n = 30) was used to examine the circadian rhythmicity of pulmonary antioxidant defense. The selected exposure times to determine whether mice have a higher PM_{2.5} sensitivity when exposed during the inactive or active phase were chosen based on the physical activity monitored in metabolic cages (see Fig. 1D).

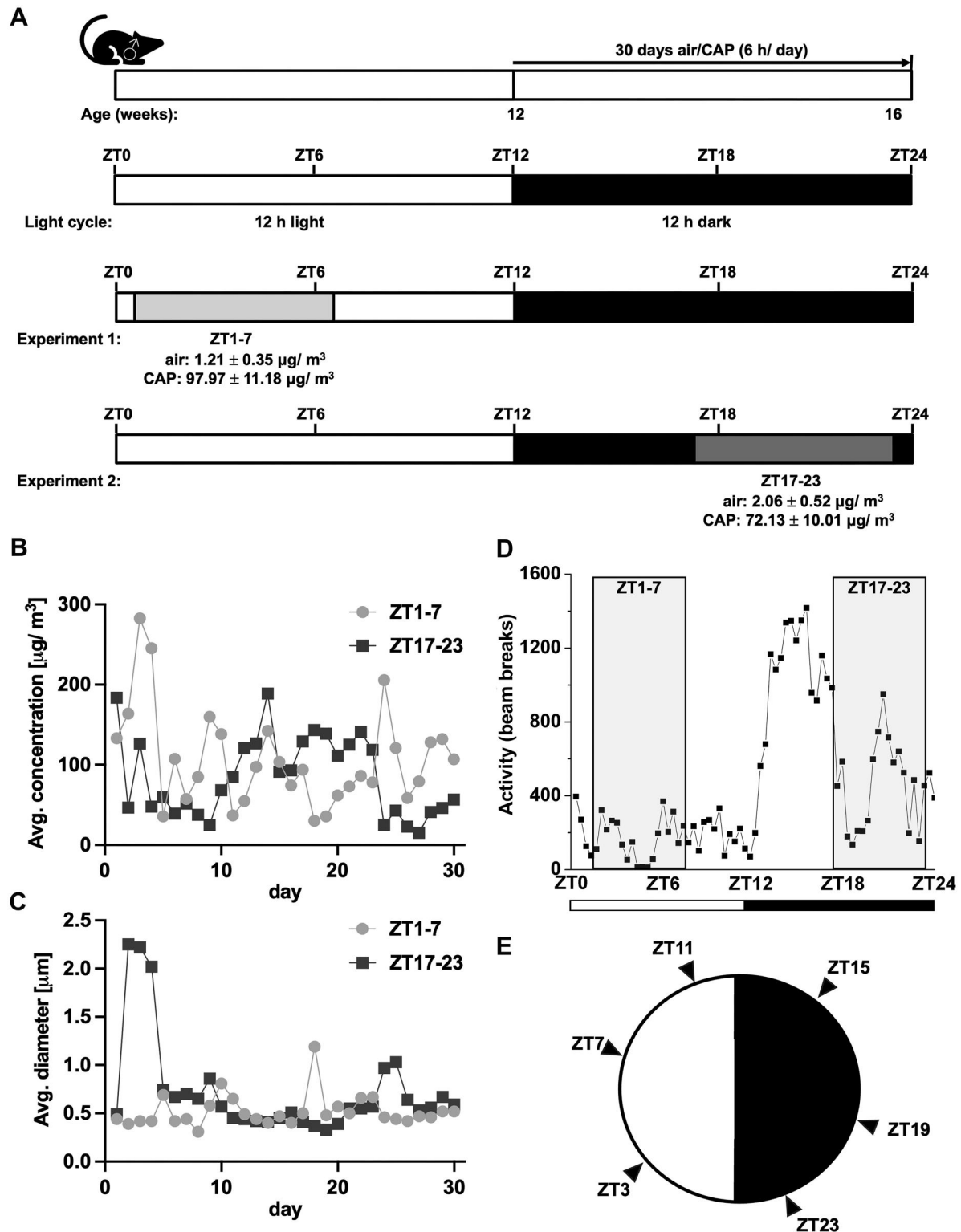


Fig. 1. Experimental design and exposure parameter. (A) Male mice (12-wks-old) maintained on a 12 h light [ZT0 to 12]:12 h dark [ZT12 to 24] (ZT = Zeitgeber in circadian time, ZT0 = beginning of light cycle, ZT12 = beginning of the dark cycle) were exposed to concentrated fine particulate matter (CAP) between either ZT1-7 (Experiment 1, air: $1.21 \pm 0.35 \mu\text{g}/\text{m}^3$, CAP: $97.97 \pm 11.18 \mu\text{g}/\text{m}^3$) or ZT17-23 (Experiment 2, air: $2.06 \pm 0.52 \mu\text{g}/\text{m}^3$, CAP: $72.13 \pm 10.01 \mu\text{g}/\text{m}^3$) for 30 consecutive days (6 h per day). Mice inhaling HEPA-filtered air were used as controls. Daily average (Avg.) real-time mass particle concentration (B) and median particle diameter (C) measured using a nephelometer. (D) Physical activity monitored over 24 h in mice using metabolic cages to determine exposure times during the inactive (light) phase [ZT1 to 7] or active (dark) phase [ZT17 to 23]. (E) Blood and tissues were collected at ZT3, 7, 11, 15, 19, and 23.

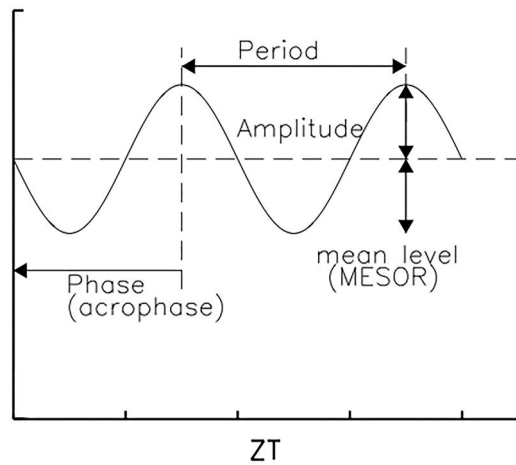


Fig. 2. Circadian rhythm analysis. Circadian rhythmicity ($P \leq 0.001$) was determined by fitting the data points to the equation: $Y = M + A \times \cos(2\pi t/\tau - \phi)$; M = mesor, A = amplitude, t = time (h), τ = period, and ϕ = acrophase.

Body weights of mice were monitored weekly during the 30 days of exposure. In addition to testing whether the exposure sensitivity depends on exposure timing (chronotoxicity), we also tested for the contingency of an outcome to the specific time of the day. For this, blood and tissues were collected from air- or CAP-inhaling mice at ZT3, 7, 11, 15, 19, and 23 ($n = 5 \times 6$ time-points, $n = 30$ in total, Fig. 1E). This not only allows to assess whether the impact of CAP exposure varies during the day; such sampling strategy also provides the necessary data points for the analysis of circadian rhythmicity (Fig. 2). Exposures in the inactive dark phase and tissue collections at ZT15, ZT19, and ZT23 were performed using a red-light source. After a 6 h fast, blood glucose levels were measured (Accu-Check glucometer, Aviva, Roche), and mice were injected with a lethal dose of pentobarbital sodium (150 mg/kg). Blood was collected by cardiac puncture using Na4-EDTA as anticoagulant, and separated plasma was snap-frozen and stored at -80°C for further biochemical analysis. Tissues were excised and weighed. Collected lungs and aortas, cleaned of periadventitial connective tissue, were snap-frozen and stored at -80°C for further analysis. All protocols for the animal experiments were approved by the University of Louisville Institutional Animal Care and Use Committee (IACUC #24448).

Assessment of glucose homeostasis

Glucose homeostasis was assessed consistently during the inactive (light) and active (dark) phase by measuring fasting blood glucose and plasma insulin level. After a 6 h fast, blood glucose levels were measured (Accu-Check glucometer, Aviva, Roche), and blood was collected for plasma isolation at ZT3, 7, 11, 15, 19, and 23. Measured fasting blood glucose and plasma insulin levels (assessed using a commercially available kit, Alpco Ltd. Chemi Rodent Insulin ELISA, Catalog number: NC0544405) were utilized to calculate HOMA-IR ($= \text{fasting blood glucose [mmol/l]} \times \text{fasting plasma insulin levels [mU/l]} / 22.5$) and HOMA- β ($= 20 \times \text{fasting plasma insulin levels [mU/l]} / \text{fasting blood glucose [mmol/l]} - 3.5$, in %) scores as markers of systemic insulin resistance or pancreatic β -cell dysfunction, respectively.

Quantitative real-time-PCR

Lungs and aortas were used to perform mRNA analysis by quantitative real-time (qRT)-PCR as described previously (Haberzettl

et al. 2016a, 2016b). Primer sets listed in Table 1 were used to measure the mRNA abundance. As an indicator for CAP-induced oxidative stress, mRNA levels of the antioxidant enzymes soluble superoxide dismutase 1 (Sod1), mitochondrial superoxide dismutase 2 (Sod2), extracellular superoxide dismutase 3 (Sod3), catalase (Cat), heme oxygenase-1 (Hmox1), glutathione transferase a, m, and p (Gsta, m, and p), and the antioxidant enzyme transcription factor of nuclear factor-erythroid factor 2-related factor2 (Nrf2) were determined. The abundance of the pro-inflammatory cytokine and chemokine mRNAs tumor necrosis factor α (TNF- α), interleukin 1 β (Il1 β), interleukin 6 (Il6), chemokine (C-C motif) ligand 2 (Ccl2, aka monocyte chemoattractant protein 1, Mcp-1), and chemokine (C-C motif) ligand 3 (Ccl3, aka macrophage inflammatory protein-1- α , Mip-1- α) were measured as common markers of inflammation that are known to be induced by PM_{2.5} exposure (Haberzettl et al. 2014, 2016a; Yue et al. 2019). Glyceraldehyde-3-phosphate dehydrogenase (Gapdh) was used as housekeeping gene, as neither CAP-exposure nor harvest time impacted the consistency of the pulmonary or aortic Gapdh mRNA abundance (data not shown). The qRT-PCR data were analyzed using the $2^{-\Delta\Delta C_t}$ method (Kurrasch et al. 2004). The cycle threshold (C_t) of the housekeeping gene, and the gene of interest was used to calculate ΔC_t [$\Delta C_t = C_t$ (target gene) $- C_t$ (housekeeping gene, Gapdh)] that accounts for variations in sample input. To identify CAP-induced changes, the data were normalized to the respective air controls ($\Delta\Delta C_t = \Delta C_t$ (CAP) $- \Delta C_t$ (air), Fold change $= 2^{-\Delta\Delta C_t}$), and data are presented as the fold change of the air controls. To identify diurnal changes, data of each timepoint were normalized to ZT3 [$\Delta\Delta C_t = \Delta C_t$ (ZT7, 11, 15, 19, or 23) $- \Delta C_t$ (ZT3), Fold change $= 2^{-\Delta\Delta C_t}$], and data are presented as the fold change of ZT3.

Plasma and lung biochemistry

Redox changes in lungs and plasma were assessed using commercially available kits following the manufacturers' guidelines. The levels of the lipoperoxidation products, thiobarbituric acid reactive substances (TBARS, ZeptoMetrix, TBARS Assay Kit, Catalog #0801192), were quantified in plasma and lungs to determine systemic and pulmonary redox changes. To determine further pulmonary redox changes, a GSH assay kit (Glutathione Colorimetric Detection Kit, Catalog #E1AGSHC) was used to detect reduced (GSH) and oxidized (GSSG) glutathione in the collected lung tissue, and the ratio of GSH: GSSG [$\text{GSH} - (2 \times \text{GSSG}) / \text{GSSG}$] was calculated. The measurement of the nitric oxide breakdown product nitrite (NO $_x$, Griess assay, Promega, Catalog #G2930) in plasma was used as a marker for systemic nitrosative stress.

Data analysis

The software Graph Pad Prism (Version 11.0.0) was used for statistical analysis and graphing of the data. Experimental data are presented as mean $\pm$ SE. To identify CAP-induced changes that occur either during the exposure in the active or inactive phase, we compared the data obtained from the CAP-exposed mice in Experiment 1 and Experiment 2 with the pooled air controls. For this, a two-way analysis of variance (ANOVA) followed by a Bonferroni's multiple comparisons test was used to determine statistical significance ($P \leq 0.05$).

The cosinor method (see Fig. 2) was used to test for the circadian rhythmicity of the diurnal pulmonary antioxidant defense enzyme mRNA abundance (Halberg et al. 1965; Refinetti et al. 2007). This method is appropriate for non-equidistant as well as serially independent data. Using the cosinor method, we compared the qRT-PCR data acquired from the lungs of air-inhaling

Table 1. Primer sequences.

mRNA	Forward primer (5'-...-3')	Reverse primer (5'-...-3')
Sod1	GATGAAGAGAGGCATGTTGGA	TGTACGGCCAATGATGGAATG
Sod2	GCGGTCGTGTAAACCTCAT	CCAGAGCCTCGTGGTACTTC
Sod3	CTGAGGACTTCCCAGTGAC	GGTGAGGGTGTGAGAGTGT
Cat	AGCGACCAGATGAAGCAGTG	TCCGCTCTCTGTCAAAGTGTG
Hmox-1	CACGCATATACCCGCTACCT	CCAGAGTGTTCATTTCGAGA
Nrf2	CTCGCTGGAAAAAGAAGTG	CCGTCCAGGAGTTCAGAGG
Gsta	TGATTGCCGTGGCTCCATTTA	CAACGAGAAAAGCCTCTCCGT
Gstm	AGCTCACGCTATTCCGGCTG	GCTCCAAGTATTCCACCTTCAGT
Gstp	ATGCCACCATACACCATGTGC	GGGAGCTGCCCATACAGAC
Tnfa	GCATGATCCGCGACGTGG AA	AGATCCATGCCGTTGGCCAG
Il1a	CTCCATGAGCTTTGTACAAGG	TGCTGATGTACCAGTTGGGG
Il6	Qiagen Catalog # PPM03015A-200	Product # 330001
Ccl2	ATGCAGGTCCTGTCTATG	GTCTGAGGTGTTGTGGA
Ccl3	ACTGACCTGGAAGTGAATGCCTGA	ATGTGGCTACTTGGCAGCAAACAG
Gapdh	AGGTCATCCCAGAGCTGAACG	GGAGTTGCTGTTGAAGTCGCA

Forward and reverse primer sets used to measure the mRNA abundance of superoxide dismutase 1, 2, and 3 (*Sod1*, 2, 3), catalase (*Cat*), heme oxygenase 1 (*Hmox1*), nuclear factor erythroid 2-related factor 2 (*Nrf2*), glutathione transferase a, m, and p (*Gsta*, m, p), tumor necrosis factor α (*Tnfa*), interleukin 1 β (*Il1b*) and 6 (*Il6*), chemokine (C-C motif) ligand 2 (*Ccl2*) and 3 (*Ccl3*), and glyceraldehyde-3-phosphate dehydrogenase (*Gapdh*).

mice that were collected during 24 h (Fig. 1E) with a cosine model of the following form: $Y = M + A \times \cos(2\pi/\tau - \phi)$. To find the best model, we varied the midline estimating statistic of rhythm (MESOR, *M*), the amplitude (*A*), period (τ), and the acrophase (ϕ). The fitting of cosine functions was performed with the Python package *lmfit* (Python package *lmfit*, version 1.3.4). The *lmfit* routine uses a Levenberg–Marquardt non-linear least-squares minimization approach to find the best-fitting model. Statistical significances were calculated using F-distributions (Refinetti et al. 2007). For statistical analyses, we employed the collection of mathematical algorithms and convenience functions for scientific computing in Python SciPy (Python *scipy*, version 1.16.3).

Results

Exposure timing determines the impact of PM_{2.5} inhalation on glucose homeostasis

To test whether exposure timing impacts CAP-induced effects on physiological, vascular, and pulmonary outcomes, mice were exposed during either the inactive light (ZT1–7) or the active dark (ZT17–23) phase for 6 h per day for 30 days. Body weight (Fig. 3A) measured weekly during the exposure experiments showed no differences between air- and CAP-inhaling mice. At the end of the experiments, fasting blood glucose levels (Fig. 3B), plasma insulin concentrations (Fig. 3C), and body weight (Fig. 3D) were determined at ZT3, 7, 11, 15, 19, and 23. In line with previous observations (Haberzettl et al. 2016b; Ribble et al. 2023), CAP exposure during the inactive phase had no impact on body weights or any measured tissue weights (Table 2), and as no differences in blood glucose or plasma insulin levels between CAP- and air-inhaling mice were observed, calculated HOMA-IR (Fig. 3E) and HOMA- β (Fig. 3F) scores were unchanged. In contrast, CAP exposure during the active phase increased blood glucose levels at ZT3 that led to an increase in both HOMA-IR and HOMA- β scores. Although CAP exposure during the active phase impaired systemic insulin sensitivity and pancreatic β -cell function at ZT3, no changes were observed at ZT7, 11, 15, 19, or 23. The observed changes in glucose homeostasis were independent of weight gain and obesity, as CAP exposure neither impacted body weight (Fig. 3A and D) nor adipose tissue weight (Table 2).

Taken together, the observed changes in HOMA-IR and HOMA- β scores at ZT3 after active-phase exposure suggest that PM_{2.5} shows chronotoxicity with a higher CAP exposure

sensitivity to develop insulin resistance (Fig. 3E) and pancreatic β -cell dysfunction (Fig. 3F) when the exposure occurred during the active phase. In addition, the impact of CAP exposure on glucose homeostasis seems to be contingent on the time of the day the measurement was taken. Changes in blood glucose levels occur only at ZT3 but not ZT7, 11, 15, 19, or 23, supporting the importance of considerations concerning the time when the measurements are taken.

Exposure timing is an important determinant of PM_{2.5}-induced vascular oxidative stress and inflammatory responses

Although our previous studies indicate that short-term PM_{2.5} exposures during the inactive phase do not impact glucose homeostasis independent of diet-induced obesity or light-induced circadian dyssynchrony (Haberzettl et al. 2016b; Ribble et al. 2023), we and others have demonstrated that PM_{2.5} exposure independently induces inflammation and oxidative stress in the blood vessels (Sun et al. 2005; Kampfrath et al. 2011; Haberzettl et al. 2016a, 2016b; Hill et al. 2021). Although its contribution to the development of cardiometabolic disease has been extensively examined, it has not been determined whether the effects of PM_{2.5} on vascular health are dependent on the exposure timing. Consequently, we investigated whether the induction of vascular oxidative stress (Fig. 4A) and inflammation (Fig. 4B) by PM_{2.5} depends on the time of the day during which the exposure occurs. To assess vascular oxidative stress and inflammation, aortas isolated from mice exposed either during the inactive or active phase were collected at ZT3 (i), 7 (ii), 11 (iii), 15 (iv), 19 (v), and 23 (vi), and the mRNA abundance of antioxidant enzymes (Fig. 4A) and cytokines/chemokines (Fig. 4B) was measured by qRT-PCR. Although exposure during the inactive phase increased the abundance of several antioxidant enzyme and cytokine mRNAs when compared with air-inhaling mice, exposure during the active phase appears to induce less vascular oxidative stress and inflammation. For instance, at ZT3 (i), CAP exposure increased the mRNA abundance of the antioxidant enzymes *Sod1*, *Sod3* and of the antioxidant enzyme transcription factor *Nrf2*, as well as the cytokines *Tnfa*, *Il1b*, and the chemokine *Ccl2* in mice exposed to CAP during the inactive phase. In contrast, at ZT3, none of the measured antioxidant enzyme or cytokine/chemokine mRNA levels were elevated in mice exposed to CAP during the active phase. This suggests a higher vascular

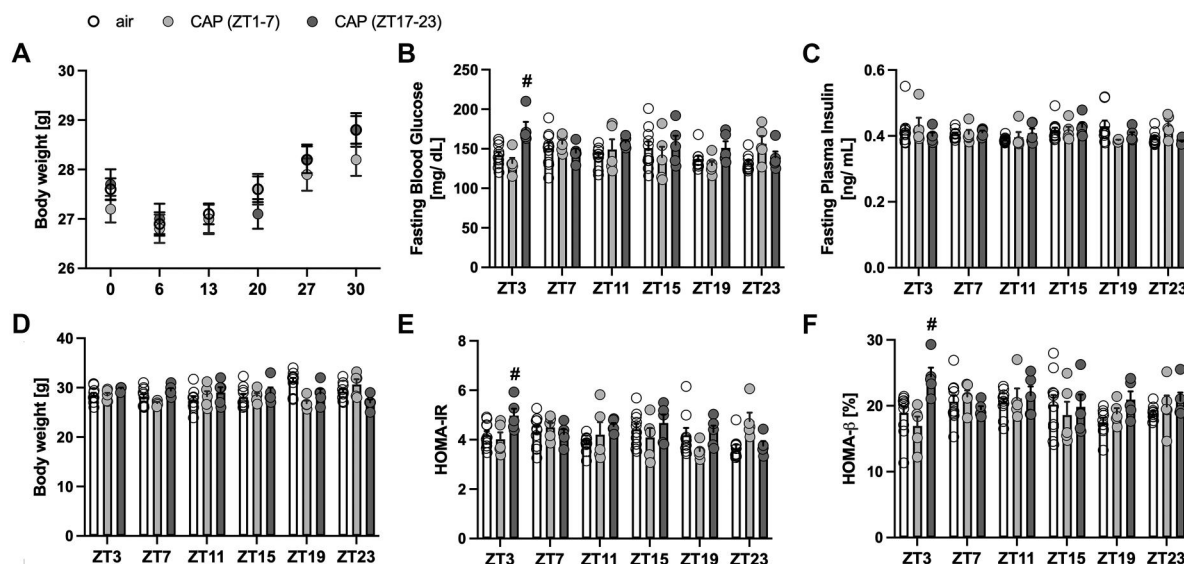


Fig. 3. Exposure timing and time-of-day dependency of the effects of PM_{2.5} on glucose homeostasis. Mice were exposed to concentrated fine particulate matter (CAP) during either the inactive (ZT1 to 7, Experiment 1, $n = 30$) or the active dark (ZT17 to 23, Experiment 2, $n = 30$) phase. During either exposure protocol mice inhaling filtered air (Experiments 1 and 2, $n = 59$ after the experimental loss of 1 animal) were used as controls. (A) Body weight measured in mice at days 0, 6, 13, 20, 27, and 30 of the exposures. Data are the mean $\pm$ SE. Fasting blood glucose (B) and plasma insulin (C) levels as well as final body weights (D) measured at ZT3, 7, 11, 15, 19, and 23. (E) HOMA-IR (= fasting blood glucose [mmol/L] $\times$ fasting plasma insulin levels [mU/L]/22.5) and (F) HOMA- β (= $20 \times$ fasting plasma insulin levels [mU/L]/fasting blood glucose [mmol/L] - 3.5, in %) scores were calculated from the measured parameter. Data are presented as single data points (CAP exposures during ZT1-7 or ZT17-23: $n = 5$ per collection time point; air: $n = 10$ per collection time point, $n = 9$ at ZT19 after the experimental loss of 1 animal) and mean $\pm$ SE, * $P \leq 0.05$ (two-way analysis of variance [ANOVA] followed by a Bonferroni's multiple comparisons) CAP (ZT17 to 23) versus air.

Table 2. Physiological parameter.

	Air	CAP (ZT1-7)	CAP (ZT17-23)
Body weight (BW)	28.8 $\pm$ 3.7	28.2 $\pm$ 0.4	28.8 $\pm$ 0.3
Lung: BW (%)	0.48 $\pm$ 0.06	0.49 $\pm$ 0.02	0.47 $\pm$ 0.02
Heart: BW (%)	0.49 $\pm$ 0.06	0.47 $\pm$ 0.01	0.48 $\pm$ 0.01
Liver: BW (%)	4.40 $\pm$ 0.57	4.59 $\pm$ 0.09	4.22 $\pm$ 0.08
Adipose: BW (%)	1.22 $\pm$ 0.16	1.40 $\pm$ 0.27	1.13 $\pm$ 0.05
Spleen: BW (%)	0.31 $\pm$ 0.04	0.33 $\pm$ 0.01	0.33 $\pm$ 0.01
Kidney: BW (%)	1.21 $\pm$ 0.16	1.23 $\pm$ 0.02	1.19 $\pm$ 0.03

Body weight (BW), and tissue: BW measured in mice exposed to concentrated fine particulate matter (CAP) during either the inactive (ZT1 to 7, Experiment 1, $n = 30$) or the active (ZT17-23, Experiment 2, $n = 30$) phase. During either exposure protocol mice inhaling filtered air (Experiments 1 and 2, $n = 59$, 1 missing data set due to an experimental loss) were used as controls. Data are mean $\pm$ SE.

susceptibility for CAP exposures during the inactive phase, as similarly observed at other time points. As such, CAP exposure during the inactive phase increased the levels of *Sod2*, *Gstp*, and *Il1 β* at ZT7 (ii), the concentrations of *Tnfa* and *Il1 β* mRNA at ZT11 (iii), and the abundance of *Sod1*, *Cat*, *Hmox1*, and *Gsta* at ZT23 (vi), whereas CAP exposure during the active phase did not increase the mRNA abundance of antioxidant enzymes or inflammatory cytokines/chemokines at these time points. The mRNA expression at ZT15 (iv) showed a more diverse pattern between inactive and active-phase exposure. As such, exposure during the inactive phase increased *Sod1* levels, whereas exposure during the active phase increased *Hmox1* abundance. At ZT19, CAP exposure during the inactive phase did not increase any antioxidant enzyme mRNA level but augmented *Ccl2* mRNA levels (v). Exposure during the active phase increased the abundance of *Il6* mRNA and decreased the levels of *Sod2*, *Cat*, and *Gsta* at ZT19 (v). In addition, we found that levels of *Hmox1* and *Gstm* at ZT7 (ii) were decreased by CAP exposure during the inactive phase, whereas

CAP exposure during the active phase decreased *Hmox1* at ZT11 (iii) and *Sod3* at ZT23 (vi).

In summary, our findings show that inactive-phase CAP exposures increased the abundance of antioxidant enzyme and cytokine/chemokine mRNA more than exposure during the active phase. This suggests a higher vascular sensitivity for CAP-induced oxidative stress and inflammation during the inactive phase. In addition to the observed chronotoxicity, it is of note that the detection of the outcomes appears to depend on the time of the day the measurements were taken, as the impact of PM_{2.5} appears to be different when measured at ZT3, 7, 11, 15, 19, or 23.

Exposure timing is an important determinant of PM_{2.5}-induced pulmonary oxidative stress and inflammatory responses

The induction of oxidative stress and inflammation in the lungs by inhaled PM_{2.5} is a long-known aspect of particle toxicology and has been found to be associated with adverse cardiovascular and metabolic health effects (Lodovici and Bigagli 2011; Haberzettl et al. 2014, 2016b, 2018; Kurlawala et al. 2023; Marchini 2023; Ribble et al. 2023). Hence, we tested for changes in the pulmonary mRNA abundance of antioxidant enzymes (Fig. 5A), as indicators of oxidative stress, and cytokines/chemokines (Fig. 5B), as markers of inflammation, in the lungs of mice exposed to CAP during either the inactive or the active phase at ZT3 (i), 7 (ii), 11 (iii), 15 (iv), 19 (v), and 23 (vi) by qRT-PCR. Both CAP exposure during the inactive and active phases increased pulmonary antioxidant enzyme, as well as cytokine and chemokine mRNA levels when compared with air-inhaling mice. However, the patterns of PM_{2.5}-induced oxidative stress and inflammation seem to have distinctive differences between the exposure during the inactive and active phases. As such, the

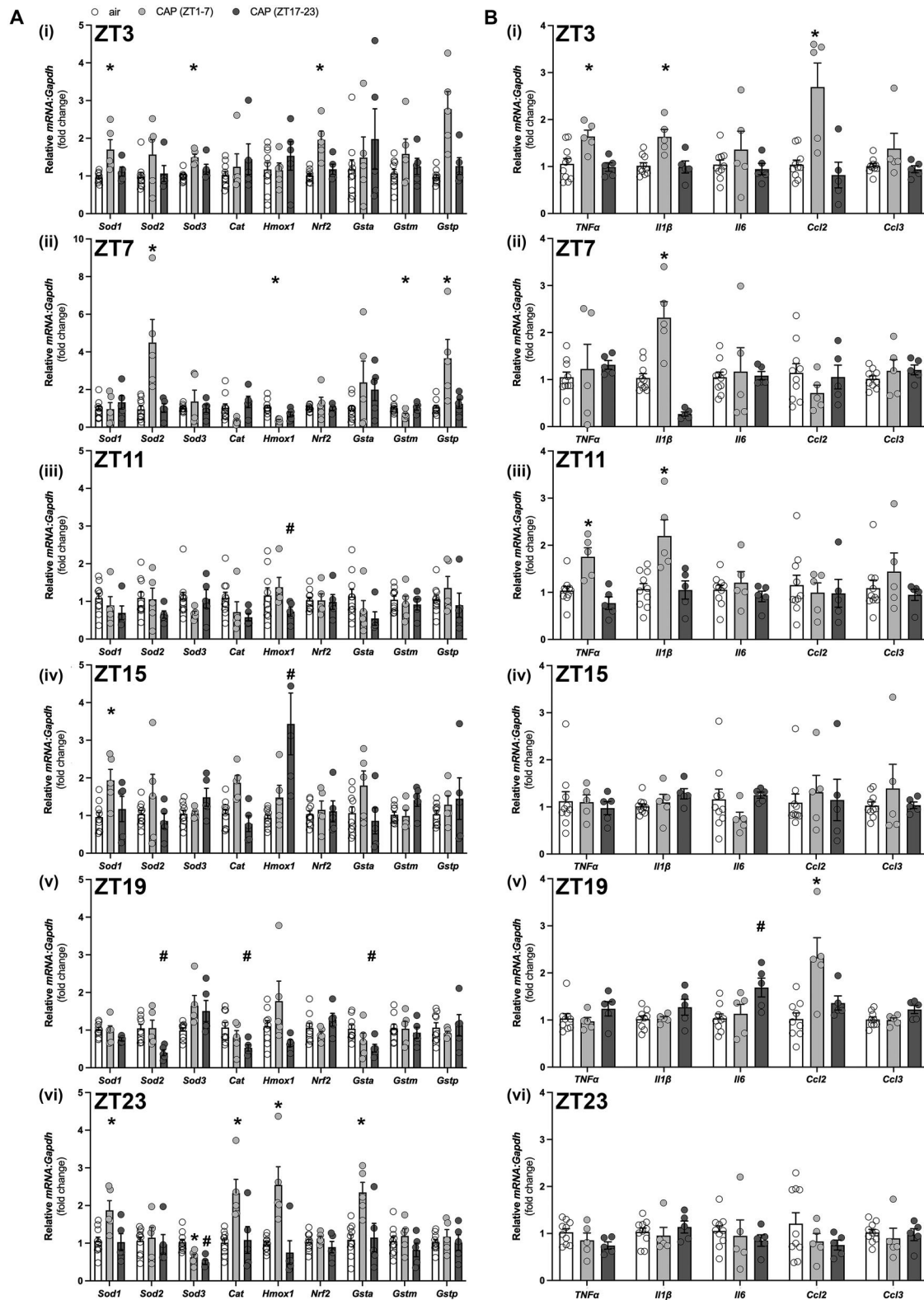


Fig. 4. Exposure timing and time-of-day dependency of PM_{2.5}-induced vascular oxidative stress and inflammatory responses. Mice were exposed during either the inactive (ZT1 to 7, Experiment 1, $n = 30$) or the active (ZT17 to 23, Experiment 2, $n = 30$) phase to CAP for 30 days. During either exposure protocol, mice inhaling filtered air (Experiments 1 and 2 combined, $n = 60$) were used as controls. Aortic mRNA levels of (A) antioxidant enzymes (superoxide dismutase 1, 2, and 3, *Sod1*, 2, and 3; catalase, *Cat*; heme oxygenase 1, *Hmox1*; nuclear factor-erythroid factor 2-related factor, 2*Nrf2*; glutathione transferase α , μ , and π , *Gsta*, μ , and π) and (B) inflammatory cytokine/chemokines (tumor necrosis factor α , *Tnf α* ; interleukin 1 β and 6, *Il1 β* and 6; chemokine [C-C motif] ligand 2 and 3, *Ccl2* and 3) determined by qRT-PCR in blood vessels collected at ZT3 (i), 7 (ii), 11 (iii), 15 (iv), 19 (v), and 23 (vi). Data are presented as single data points and mean $\pm$ SE. A two-way analysis of variance (ANOVA) followed by a Bonferroni's multiple comparisons test was used to determine statistical significance * $P \leq 0.05$ CAP (ZT1 to 7, $n = 5$ per collection time point), # $P \leq 0.05$ CAP (ZT17 to 23, $n = 5$ per collection time point) versus air ($n = 10$ per collection time point, $n = 9$ at ZT19 after the experimental loss of 1 animal).

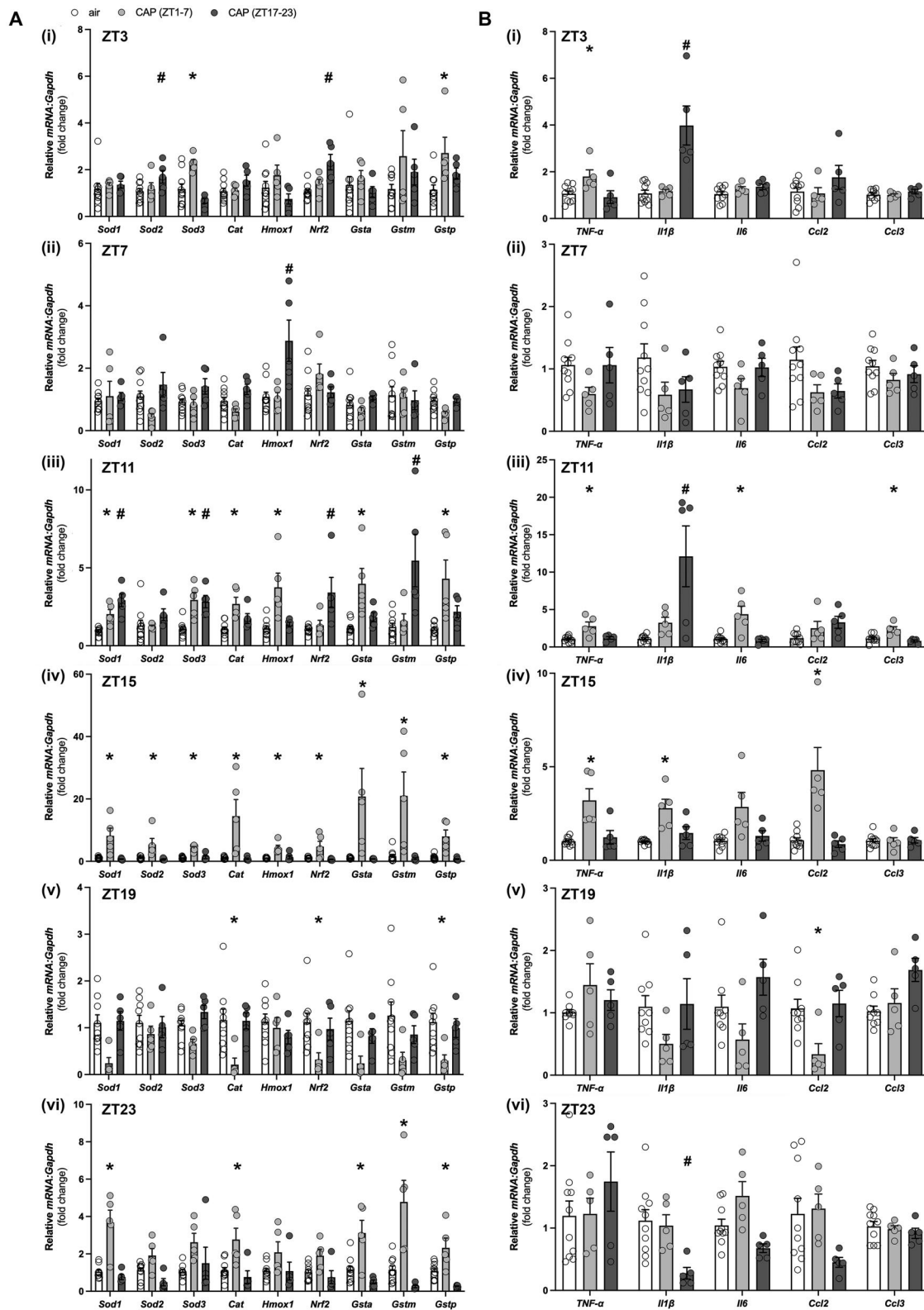


Fig. 5. Exposure timing and time-of-day dependency of PM_{2.5}-induced pulmonary oxidative stress and inflammatory responses. Mice were exposed during either the inactive (ZT1 to 7, Experiment 1, $n = 30$) or the active (ZT17 to 23, Experiment 2, $n = 30$) phase to CAP for 30 days. During either exposure protocol mice inhaling filtered air (Experiments 1 and 2 combined, $n = 60$) were used as controls. Pulmonary mRNA levels of (A) antioxidant enzymes (superoxide dismutase 1, 2, and 3, *Sod1*, 2, and 3; catalase, *Cat*; heme oxygenase 1, *Hmox1*; nuclear factor-erythroid factor 2-related factor, 2*Nrf2*; glutathione transferase α , m and p , *Gst a*, m , and p) and (B) inflammatory cytokine/chemokines (tumor necrosis factor α , *Tnf α* ; interleukin 1 β and 6, *Il1 β* and 6; chemokine (C-C motif) ligand 2 and 3, *Ccl2* and 3) determined by qRT-PCR in lungs collected at ZT3 (i), 7 (ii), 11 (iii), 15 (iv), 19 (v), and 23 (vi). Data are presented as single data points and mean $\pm$ SE. A two-way analysis of variance (ANOVA) followed by a Bonferroni's multiple comparisons test was used to determine statistical significance * $P \leq 0.05$ CAP (ZT1 to 7, $n = 5$ per collection time point), # $P \leq 0.05$ CAP (ZT17 to 23, $n = 5$ per collection time point) versus air ($n = 10$ per collection time point, $n = 9$ at ZT19 after the experimental loss of 1 animal).

impact of CAP exposure seems to have a stronger impact when occurring during the inactive phase. For instance, although CAP exposure during the active phase had no effect on the mRNA abundance of any antioxidant enzyme measured at ZT15 (iv), CAP exposure during the inactive phase significantly increased the abundance of the antioxidant enzymes *Sod1*, 2, and 3, *Cat*, *Hmox1*, the transcription factor *Nrf2*, as well as *Gsta*, *m*, and *p*. In addition, mRNA levels of the cytokines *Tnfa* and *Il1b* were elevated in mice exposed to CAP during the inactive phase but not in active-phase CAP-exposed mice. Similarly, at ZT23 (vi), only CAP exposure during the inactive phase increased the levels of antioxidant enzyme mRNAs, namely *Sod1*, *Cat*, and *Gsta*, *m*, and *p*. Yet, at ZT3 (i) and ZT11 (iii), the qRT-PCR results appeared to be more diverse. As such, inactive phase exposure increased the mRNA abundance of *Sod3*, *Gstp*, and *Tnfa* at ZT3 and *Sod1*, *Sod3*, *Cat*, *Hmox1*, *Gsta*, *Gstp*, *Tnfa*, and *Il6* at ZT11, whereas active-phase exposure increased *Sod2*, *Nrf2*, and *Il1b* mRNA abundance at ZT3 and *Sod1*, *Sod3*, *Nrf2*, *GSTM*, and *Il1b* mRNA at ZT11, indicating that exposure during the active and inactive phases induce a more comparable oxidative stress and inflammatory response during the inactive phase in the early morning (ZT3) and late afternoon (ZT11). At ZT7, only *Hmox1* mRNA was elevated in the lungs of active-phase CAP-exposed mice (ii). However, measured mRNA levels seem to show more variance at ZT7, as similarly observed at ZT19 (v), a time point at which exposure neither during the inactive nor during the active phase induced an increase in mRNA levels of antioxidant or inflammatory markers. Yet, exposure during the inactive phase decreased the mRNA abundance of *Cat*, *Nrf2*, *Gstp*, and *Ccl2* at ZT19.

As in the blood vessels, CAP exposure during the inactive phase appears to induce different pulmonary mRNA level changes than exposures during the active phase, supporting the

idea of chronotoxicity. Although more complex than the findings in the aorta, CAP exposure during the inactive phase seems to induce a stronger pulmonary oxidative stress and inflammatory response than active-phase exposure. However, as in the aorta, the observed pulmonary changes depended on the time of the day when the measurements were taken. As such, our results indicate that both exposure windows induce an oxidative stress and inflammatory response in the early morning (ZT3) and late afternoon (ZT11). However, the expression pattern differed between both exposure windows.

Exposure timing is an important determinant of PM_{2.5}-induced pulmonary and systemic oxidative stress

Research results from our and other laboratories indicate that PM_{2.5} triggers cardiovascular and metabolic toxicity, at least in part, by the induction of pulmonary oxidative stress that traverses systemically (Kampfrath et al. 2011; Haberzettl et al. 2014, 2016b, 2018; Li et al. 2016; Strak et al. 2017; Riggs et al. 2020; Kurlawala et al. 2023; Ribble et al. 2023; Kitching et al. 2025; Munzel et al. 2025). Oxidative stress, characterized by an imbalance between the production of reactive oxygen species (ROS) and pulmonary antioxidant defense, results, among other things, in the depletion of glutathione and the formation of lipid peroxidation products in the lungs. The PM_{2.5}-induced increase in pulmonary oxidative stress can then spread systemically, leading to an increase in circulating lipid peroxidation products and a decrease in plasma NO_x (Cipak Gasparovic et al. 2017). Hence, to test for CAP-induced pulmonary and systemic redox changes, we measured TBARS as a marker for lipid peroxidation in the lungs (Fig. 6A, i) and plasma (Fig. 6A, ii), measured circulating NO_x levels (Fig. 6A, iii), and examined the depletion of GSH (Fig. 6B, ii) by

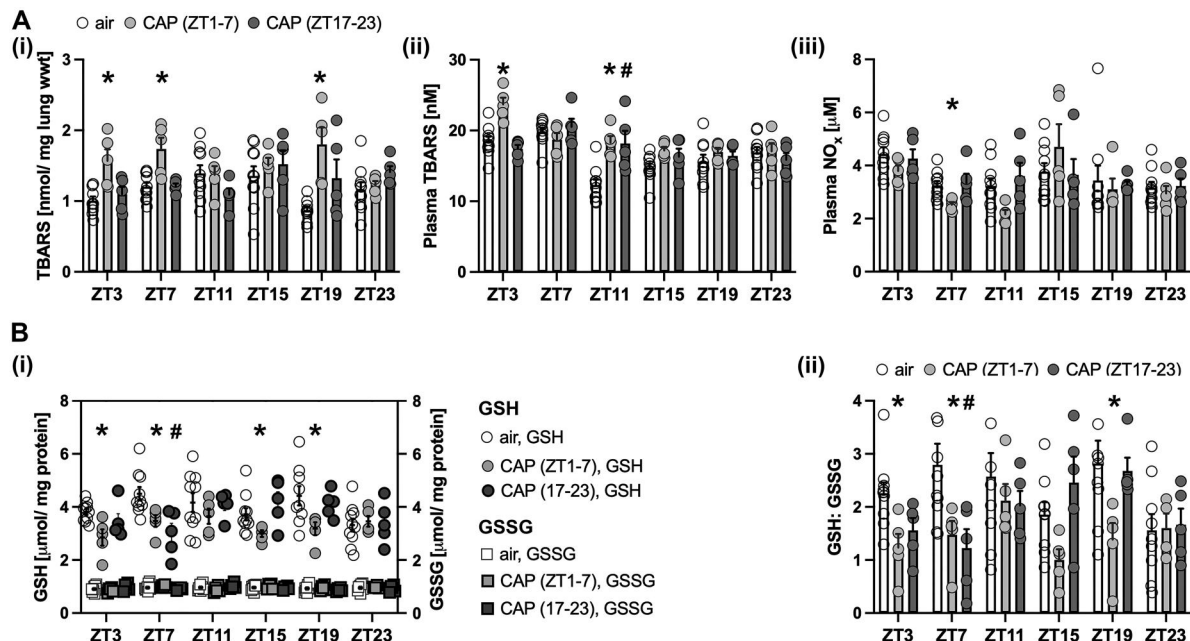


Fig. 6. Exposure timing and time-of-day dependency of PM_{2.5}-induced pulmonary and systemic redox changes. Mice were exposed during either the inactive (ZT1 to 7, Experiment 1, *n* = 30) or the active (ZT17 to 23, Experiment 2, *n* = 30) phase to CAP for 30 days. During either exposure protocol mice inhaling filtered air (Experiments 1 and 2 combined, *n* = 60) were used as controls. Lungs and blood for plasma isolation were collected at ZT3, 7, 11, 15, 19, and 23. (A) Pulmonary (i) and circulating (ii) thiobarbituric acid reactive substances (TBARS) concentration and plasma nitrite (NO_x) levels (iii) as well as (B) amounts of pulmonary reduced glutathione (GSH) and (oxidized) glutathione disulfide (GSSG, i) or GSH:GSSG ratios (ii). Data are presented as single data points and mean ± SE. A two-way analysis of variance (ANOVA) followed by a Bonferroni's multiple comparisons test was used to determine statistical significance **P* ≤ 0.05 CAP (ZT1 to 7, *n* = 5 per collection time point), **P* ≤ 0.05 CAP (ZT17 to 23, *n* = 5 per collection time point) versus air (*n* = 10 per collection time point, *n* = 9 at ZT19 after the experimental loss of 1 animal).

quantifying GSH and GSSG concentrations (Fig. 6B, i) in the lungs. In mice exposed during the inactive phase, CAP exposure increased pulmonary (Fig. 6A, i) and circulating (Fig. 6A, ii) TBARS levels at ZT3, 7, and 19 or at ZT3 and 11, respectively, and decreased plasma NO_x levels at ZT7 (Fig. 6A, iii). In contrast, exposure to concentrated PM_{2.5} during the active phase had no effect on pulmonary TBARS levels (Fig. 6A, i) or circulating NO_x concentration (Fig. 6A, iii) and only increased plasma TBARS levels at ZT11 (Fig. 6A, ii). Similarly, the impact of CAP on pulmonary GSH levels (Fig. 6B) appeared to be more severe when the exposure occurred during the inactive phase. Inactive-phase exposure induced a decrease in GSH concentrations at ZT3, 7, 15, and 19 (Fig. 6B, i) that significantly impacted the GSH: GSSG ratio at ZT3, 7, and 19 (Fig. 6B, ii), whereas active-phase exposure had only an effect at ZT7.

Taken together, these results indicate that mice that are exposed during their active phase have a better protection against CAP-induced pulmonary and systemic oxidative stress.

Circadian rhythm of pulmonary antioxidant defense mechanism

Because of the increasing evidence that antioxidant defense mechanisms show circadian pattern (Blanco et al. 2007; Jang et al. 2011; Xu et al. 2012; Wilking et al. 2013; Pekovic-Vaughan et al. 2014), we measured changes in the pulmonary antioxidant enzyme mRNA abundance at ZT3, 7, 11, 15, 19, and 23 by qRT-PCR and tested for their circadian rhythmicity. Our 24 h measurements showed that the levels of antioxidant enzyme mRNAs were elevated during the active phase and highest at ZT19 (Table 3). As such, we found that relative to ZT3, at ZT19 the mRNA abundance of *Sod1*, *Sod2*, *Sod3*, *Cat*, *Hmx1*, *Nrf2*, *Gsta*, *Gstm*, and *Gstp* were about 3 to 30-fold higher than at ZT3. Using the cosinor method (see Fig. 2) changes in amplitude (A), midline estimating statistic of rhythm (MESOR), period (τ), and acrophase (φ) were determined (Fig. 7, Table 4). The cosinor analysis, yielding P value ≤0.001 and period (τ) durations ranging from 19.64 ± 1.21 h to 26.43 ± 8.31 h indicates a circadian rhythmicity in the mRNA abundance of all examined antioxidant enzymes. Interestingly, *Sod1*, 2, and 3, *Cat*, *Hmx1*, *Nrf2*, and *Gsta*, *m* and *p* showed a maximum within the active phase. For instance, *Cat* showed an amplitude of 8.91 ± 3.07-fold, a MESOR of 6.91 ± 2.01-fold, and a peak time (acrophase) of φ = 20.98 ± 2.25 h, indicating that a maximal expression of *Cat* can be found during the active phase around ZT21. Similarly, peaked the mRNA abundance of *Sod2*, *Hmx1*, and *Nrf2* during the active phase at ZT18, levels of *Gstm* and *p* were highest at ZT20, and *Sod1* mRNA concentrations spiked around ZT22.

Discussion

Results of this study demonstrate that PM_{2.5} toxicity depends on the time during which the exposure occurs, and that the observed effects were contingent on the time-of-measurement. The observed variation in metabolic, vascular, and pulmonary sensitivity after exposure during either the inactive or active phase supports the idea of PM_{2.5} chronotoxicity. As such, our data show that exposure to concentrated PM_{2.5} during the inactive phase induces stronger oxidative stress and inflammatory responses in the aorta (Fig. 4) and lungs (Fig. 5) than exposure during the active phase. Moreover, inhalation of CAP during the inactive phase, a time at which the pulmonary antioxidant defense mechanisms appear to be lower (Fig. 7), induced stronger pulmonary and systemic redox changes, as evident by a more severe induction of lipid peroxidation and a stronger depletion of NO_x and GSH (Fig. 6). Although inhalation of concentrated PM_{2.5} during the inactive phase was more potent to induce vascular and pulmonary toxicity, exposure during the active phase, when antioxidant defense seems to peak, affected glucose homeostasis. These data suggest that CAP exposure during the inactive phase is more prone to induce vascular and pulmonary toxicity, an effect that is possibly promoted by a less effective pulmonary antioxidant defense. In contrast, exposure during the active phase induced changes in glucose homeostasis at ZT3. Although exposure during the active phase increases systemic glucose levels only at ZT3, changes in glucose homeostasis appear regardless of the stronger ability of the mouse lungs to cope with CAP-induced oxidative stress.

Regardless, PM_{2.5} toxicity not only depends on the exposure timing (chronotoxicity), it seems also to be contingent on the time of the day the observation was made. As such PM_{2.5} inhalation during the active phase impacted glucose homeostasis only at ZT3. At ZT3, CAP exposure during the active phase impaired insulin sensitivity and β-cell function as indicated by changes in calculated HOMA-IR and HOMA-β scores that did not occur when mice were exposed during the inactive phase. Similarly dependent on the observation time were the measurements in lungs and aortas. For instance, inhalation of concentrated PM_{2.5} during the inactive phase induced the strongest vascular oxidative stress and inflammatory responses at ZT 3 and ZT23, whereas pulmonary oxidative and inflammatory stress markers were more pronounced at ZT11 and ZT15. Our findings that the sensitivity to PM_{2.5} is based on the time of day (chronotoxicity) and that PM_{2.5} toxicity shows diurnal pattern could contribute to the health impact of PM_{2.5} inhalation.

Inherently, PM_{2.5} inhalation has been found to impair cardiovascular and metabolic function that show diurnal variations and to increase the risk for cardiovascular events, which are

Table 3. Daily variations in pulmonary antioxidant enzyme mRNA abundance.

mRNA	ZT3	ZT7	ZT11	ZT15	ZT19	ZT23
<i>Sod1</i>	2.81 ± 0.64	1.73 ± 0.31	1.98 ± 0.30	3.50 ± 0.45	5.82 ± 0.89	2.64 ± 0.34
<i>Sod2</i>	1.21 ± 0.15	1.73 ± 0.24	2.03 ± 0.40	3.19 ± 0.55	5.34 ± 0.72	2.34 ± 0.38
<i>Sod3</i>	1.17 ± 0.17	0.74 ± 0.16	1.08 ± 0.16	1.00 ± 0.14	2.94 ± 0.66	1.26 ± 0.24
<i>Cat</i>	1.26 ± 0.25	4.65 ± 0.71	2.26 ± 0.39	3.21 ± 0.43	28.40 ± 11.11	2.47 ± 0.34
<i>Hmx1</i>	1.51 ± 0.37	2.11 ± 0.44	2.22 ± 0.59	2.16 ± 0.57	5.89 ± 2.31	3.46 ± 0.79
<i>Nrf2</i>	1.19 ± 0.22	2.39 ± 0.45	1.73 ± 0.22	3.07 ± 0.40	5.58 ± 1.79	2.28 ± 0.35
<i>Gsta</i>	1.36 ± 0.30	3.21 ± 0.83	2.11 ± 0.31	2.59 ± 0.53	7.64 ± 1.22	2.60 ± 0.41
<i>Gstm</i>	1.32 ± 0.26	4.36 ± 0.72	2.28 ± 0.40	2.87 ± 0.55	14.42 ± 4.88	2.94 ± 0.63
<i>Gstp</i>	1.24 ± 0.25	5.08 ± 0.95	2.51 ± 0.41	3.18 ± 0.49	20.33 ± 4.59	3.04 ± 0.70

Pulmonary mRNA levels of the antioxidant enzymes superoxide dismutase 1, 2, and 3 (*Sod1*, 2, and 3), catalase (*Cat*), heme oxygenase 1 (*Hmx1*), nuclear factor-erythroid factor 2-related factor 2 (*Nrf2*), glutathione transferase *a*, *m*, and *p* (*Gsta*, *m*, and *p*) normalized to the abundance of *Gapdh* mRNA determined by qRT-PCR in lungs collected from air-inhaling mice at ZT3, 7, 11, 15, 19, and 23. Data are the mean ± SE (n = 14 per time point).

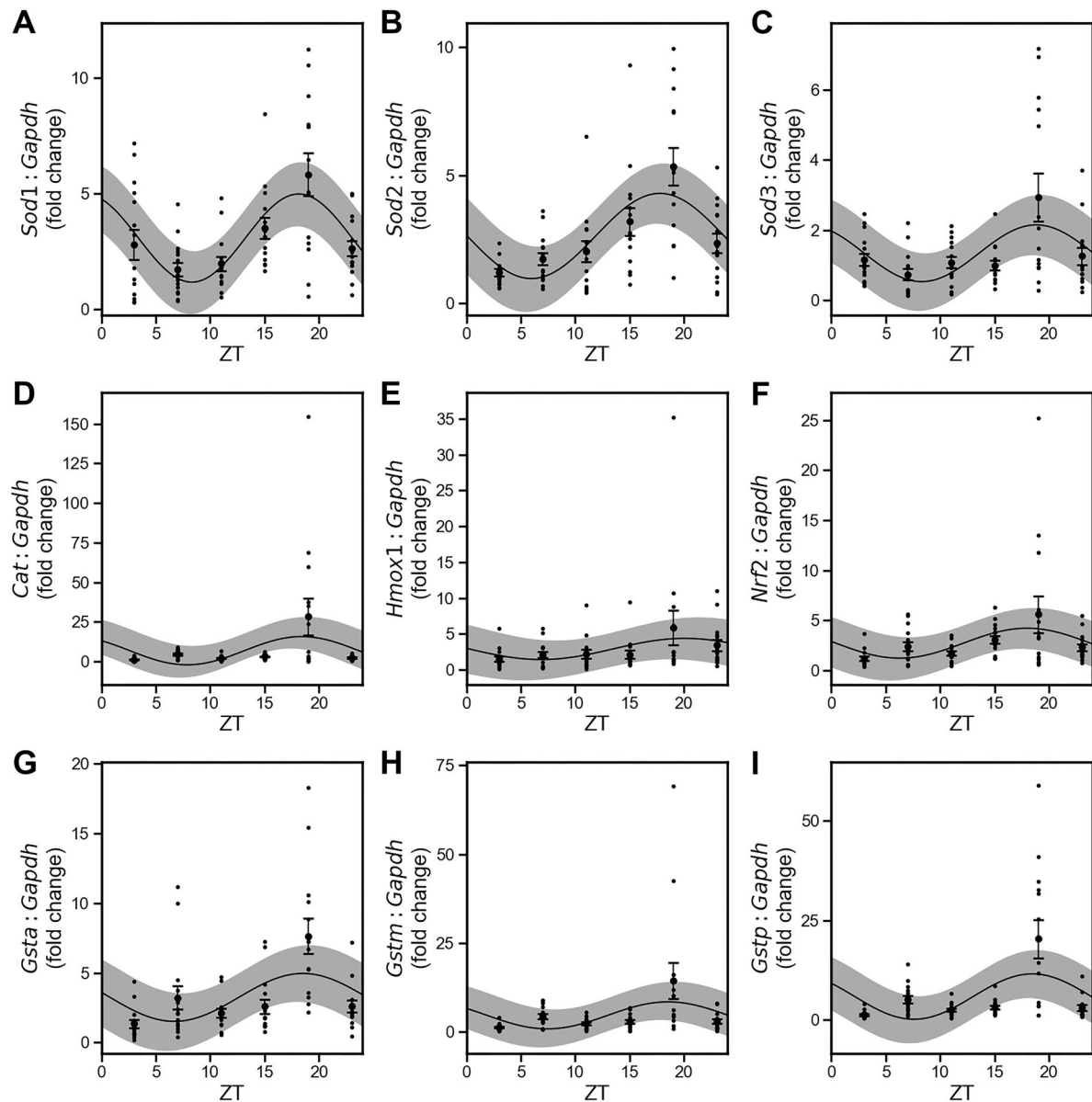


Fig. 7. Circadian rhythmicity in pulmonary antioxidant enzymes mRNA abundance. Lungs isolated from filtered air-inhaling mice were collected at ZT3, 7, 11, 15, 19, and 23 and qRT-PCR was used to determine the diurnal abundance of pulmonary mRNA of the antioxidant enzymes superoxide dismutase 1 (Sod1, A), 2 (Sod2, B) and 3 (Sod3, C), catalase (Cat, D), heme oxygenase 1 (Hmx1, E), nuclear factor-erythroid factor 2-related factor 2 (Nrf2, F), glutathione transferase a (Gsta, G), m (Gstm, H), and p (Gstp, I). Circadian rhythmicity was determined by fitting oscillations in mRNA abundance (normalized to Gapdh in fold change) using the equation: $Y = M + A \times \cos(2\pi/\tau - \phi)$ with M = mesor, A = amplitude, τ = period, and ϕ = acrophase. Data are presented as single data points, mean $\pm$ SE, and a fitted curve including a 3 σ (gray area) CI (n = 15 per time point, n = 14 at ZT19 after the experimental loss of 1 animal).

Table 4. Circadian rhythmicity in pulmonary antioxidant enzyme mRNA levels.

mRNA	Amplitude (fold change)	Mesor (fold change)	τ (h)	ϕ (h ~ ZT)	P
Sod1	1.90 $\pm$ 0.34	3.08 $\pm$ 0.23	19.64 $\pm$ 1.21	22.17 $\pm$ 1.15	$\leq 0.001^*$
Sod2	1.66 $\pm$ 0.29	2.63 $\pm$ 0.20	23.62 $\pm$ 2.20	18.08 $\pm$ 1.49	$\leq 0.001^*$
Sod3	0.81 $\pm$ 0.21	1.34 $\pm$ 0.14	21.07 $\pm$ 2.10	21.44 $\pm$ 1.68	$\leq 0.001^*$
Cat	8.91 $\pm$ 3.07	6.91 $\pm$ 2.01	21.05 $\pm$ 2.75	20.98 $\pm$ 2.25	$\leq 0.001^*$
Hmx1	1.48 $\pm$ 0.63	2.91 $\pm$ 0.45	26.43 $\pm$ 8.31	18.23 $\pm$ 3.97	$\leq 0.001^*$
Nrf2	1.50 $\pm$ 0.50	2.71 $\pm$ 0.34	23.41 $\pm$ 3.97	18.42 $\pm$ 2.71	$\leq 0.001^*$
Gsta	1.72 $\pm$ 0.49	3.24 $\pm$ 0.33	23.63 $\pm$ 3.45	18.81 $\pm$ 2.25	$\leq 0.001^*$
Gstm	3.80 $\pm$ 1.38	4.65 $\pm$ 0.91	22.18 $\pm$ 3.45	20.07 $\pm$ 2.52	$\leq 0.001^*$
Gstp	5.71 $\pm$ 1.48	5.82 $\pm$ 0.97	21.75 $\pm$ 2.30	20.37 $\pm$ 1.76	$\leq 0.001^*$

Analysis of the circadian rhythmicity of the abundance of the pulmonary antioxidant enzymes mRNA superoxide dismutase 1, 2, and 3 (Sod1, 2, and 3), catalase (Cat), heme oxygenase 1 (Hmx1), nuclear factor-erythroid factor 2-related factor 2 (Nrf2), and glutathione transferase a, m, and p (Gsta, m, and p). Levels of pulmonary antioxidant enzyme mRNA were determined by qRT-PCR in lungs collected from air-inhaling mice at ZT3, 7, 11, 15, 19, and 23 (n = 14 per time point). Values were normalized to the abundance of Gapdh mRNA. Circadian rhythmicity (*P ≤ 0.001) was determined by fitting the data points to the following equation: $Y = M + A \times \cos(2\pi t/\tau - \phi)$; M = mesor, A = amplitude, t = time (h), τ = period (h), and ϕ = acrophase.

more likely to occur at distinctive times-of-the-day (Zimmet et al. 1974; Millar-Craig et al. 1978; Schulz et al. 1983; Mejean et al. 1988; Cohen et al. 1997; Elliott 1998; Zhang and Jain 2021; Knutson et al. 2025). For instance, blood pressure expresses a 24 h pattern, and blood glucose levels, plasma insulin concentrations, glucose tolerance, as well as insulin sensitivity show diurnal variations. In humans, hyperglycemic and cardiovascular events occur more likely during the night or in the early morning hours. Because air pollution exposure has been found to impact physiological functions that occur in diurnal patterns such as insulin sensitivity and blood pressure (Brook et al. 2013; Chan et al. 2015; Xu et al. 2022; Tasmin et al. 2024) and to trigger hyperglycemic and cardiovascular events that seem to be more likely at certain time points (Kim et al. 2018; Mills and Pope 2022; Fleischer et al. 2026), exposure during different diurnal phases with different physiological attributes and event likelihoods could affect PM_{2.5} susceptibility.

Although the exact mechanism(s) by which PM adversely impacts cardiometabolic health have not been identified, several studies suggest that pulmonary as well as systemic oxidative stress contributes to the harmful effects of PM exposures (Ghio et al. 2002, 2012; Kampfrath et al. 2011; Valavanidis et al. 2013; Haberzettl et al. 2014, 2016b). Inhaled particles can induce oxidative stress directly by generating ROS due to the redox reaction of its constituents, such as quinones and metals, or indirectly by activating pro-inflammatory host defense mechanisms (Valavanidis et al. 2005; Ghio et al. 2012; Haberzettl et al. 2014). In any case, biological antioxidant defense mechanisms are in place to mitigate the induction of oxidative stress and the oxidation of biological molecules that leads to tissue injury and dysfunction. Interestingly, expanding evidence indicates that both oxidation of biomolecules as well as the abundance of antioxidant defense enzymes and molecules follow circadian rhythmicity (Blanco et al. 2007; Jang et al. 2011; Anea et al. 2012; Xu et al. 2012; Lee et al. 2013; Wilking et al. 2013; Pekovic-Vaughan et al. 2014). This suggests a tight connection between circadian rhythm and oxidative stress and that the ability to combat PM_{2.5}-induced oxidative stress depends on the time of the day (Haberzettl 2018). Considering this, the impact of PM_{2.5} is likely to be more pronounced when the levels of the antioxidant enzymes and molecules are low, and consequently, the antioxidant defense is less effective in coping with CAP-induced oxidative stress.

In the current study, we found that the pulmonary antioxidant gene expression varies during the light/dark cycle (aka, inactive/active phase) and shows a circadian pattern (Fig. 7). The lungs are the primary site of PM_{2.5} deposition (Oberdorster et al. 2005); hence, any mechanism linking PM_{2.5} exposure to systemic toxicity should establish a direct association with the lungs. Our previous study showed that constitutive overexpression of pulmonary extracellular superoxide dismutase (ecSOD) prevented PM_{2.5}-induced oxidative stress and inflammation in the lungs and blood vessels, averted systemic redox changes (i.e. restored plasma NO_x and lung GSH levels), as well as the changes in glucose homeostasis observed in light-induced circadian desynchrony (Haberzettl et al. 2016b, 2018; Hill et al. 2021; Ribble et al. 2023). Collectively, these data suggest that secondary oxidation products, generated in response to pulmonary oxidative stress, are key mediators of these vascular effects induced by PM_{2.5}. In addition, our previous study showed that female mice that have a higher pulmonary antioxidant defense were protected against the PM_{2.5}-induced cardiometabolic effects (Kitching et al. 2025). Hence, a higher potential to combat oxidative stress in the lungs during certain times of the day would protect against PM_{2.5}-

induced toxicity. In line with this notion, findings of the current study show that pulmonary and vascular oxidative stress and inflammatory responses are less severe in mice that were exposed to concentrated PM_{2.5} during the active phase when the abundance of antioxidant enzyme mRNA in the lungs is highest. In contrast, CAP exposure during the inactive phase, when the abundance of antioxidant enzyme mRNA in the lungs was in a depression, stronger vascular and pulmonary responses were observed. Our findings suggest that PM_{2.5} chronotoxicity is driven by the expression of pulmonary antioxidant enzymes that show circadian rhythmicity and that define the timely ability to protect against PM_{2.5}-induced pulmonary oxidation. Although we have not examined diurnal changes of antioxidant enzymes in the aorta, an enhanced vascular protection could contribute as well. Consequently, it would be important to investigate diurnal vascular changes in a future study. On the same notion, we found that female mice, possibly by an enhanced pulmonary antioxidant defense mechanism, are protected against metabolic and cardiovascular PM_{2.5} toxicity (Kitching et al. 2025). Therefore, we did not include female mice in this study. However, considering sex as a biological variable would be important, as the results of previous work on sex-specific effects of PM_{2.5} are still controversial (Liao et al. 2023). Because interactions between the human menstrual cycle and the mouse estrous cycle with the circadian rhythm are reported (Alvord et al. 2022; Begemann et al. 2025), it would be interesting to test whether circadian changes in antioxidant defense mechanisms drive PM_{2.5} chronotoxicity in females and how such effects are impacted by sex hormones.

Our findings also support our past research on the contribution of pulmonary oxidative stress to the cardiovascular PM_{2.5} toxicity. In contrast, the impact of PM_{2.5} on glucose homeostasis seems to be independent of pulmonary oxidative stress as it only occurs in mice exposed at the active phase that show a higher protection. Although the exposure to concentrated PM_{2.5} impacts both, HOMA-IR and HOMA- β scores, it is noteworthy that this effect is driven by an increase in blood glucose levels rather than changes in plasma insulin. In contrast, chronic exposure to concentrated PM_{2.5} increased blood glucose and plasma insulin (Palanivel et al. 2020). However, although our current study supports the idea that circadian rhythm regulates sensitivity to short-term PM_{2.5} exposure, the study by Palanivel et al. identifies the circadian rhythm as a target of chronic PM_{2.5} exposure characterized by a disruption of clock gene expression in the liver and brown adipose tissue. Although there is growing evidence that PM_{2.5} impacts the molecular regulation of the circadian rhythm (Cantone et al. 2020; Rajagopalan et al. 2020; Monti et al. 2021), data presented here support the idea that the circadian rhythm influences PM_{2.5} sensitivity. Regardless of the directionality, further research is needed to identify how the circadian rhythm impacts the sensitivity to air pollution exposure, or vice versa, that exposure to air pollution impacts the circadian clocks.

In conclusion, our results revealed time-of-day susceptibility to PM_{2.5} exposure (chronotoxicity). Because mice are exposed to real-world particles, our results bear directly on susceptibility windows for PM_{2.5} exposure in humans. Although mice are nocturnal, being active during the night and inactive during the day, our findings are applicable to human health, as there is growing evidence that antioxidant defense mechanisms are peaking in the morning during the active phase (Wilking et al. 2013; Patel et al. 2014). Based on our results showing the chronotoxicity of PM_{2.5}, translational studies could be developed to assess whether there are specific times in a day when humans are more sensitive to PM_{2.5}-induced pulmonary, cardiovascular, and metabolic

injury. Findings of such studies could result in either regulatory or preventive strategies to reduce PM_{2.5} exposure at certain susceptible times. Identifying sensitive exposure windows could help to limit the impact of PM_{2.5} on human health through discouraging air pollution exposure at certain susceptible times. Recommendations to avoid exposures during specific times might be a simple way to reduce the cardiovascular and metabolic health risk of air pollution.

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Conflicts of interest. None declared.

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